Khwaja Yunus Ali University

School of Science and Engineering

Department of Computer Science and Engineering

Final Examination: Spring Semester 2019

Program: B.Sc. in CSE (Batch: 9^{th}) 1^{st} Year 2^{nd} Semester

Course Title: Physics – II

Course Code: PHY 1201

Total Time: 90 min | Full Marks: 40

Part-A

[Write the no. of the appropriate answer of the questions sequentially] **10 1 = 10**

1. The bending of light into the geometrical shadow is called ……………
   - a) reflection.
   - b) refraction.
   - c) interference.
   - d) diffraction
   - e) polarization

2. 1eV = ?
   - a) $931 Joule$
   - b) $9.1 \times 10^{-31} Joule$
   - c) $6.63 \times 10^{-34} Joule$
   - d) $1.67 \times 10^{-27} Joule$
   - e) $1.6 \times 10^{-19} Joule$

3. What is the mirror nucleus of $^{7}_{4}Be$?
   - a) $^{4}_{2}He$
   - b) $^{7}_{3}Li$
   - c) $^{8}_{4}Be$
   - d) $^{8}_{3}Li$
   - e) $^{14}_{8}O$

4. Which nucleus have the maximum binding energy per nucleon?
   - a) Al
   - b) Xe
   - c) Fe
   - d) Au
   - e) U

5. In the nuclear reaction, proton is denoted by ………….
   - a) $^{4}_{2}He$
   - b) $^{1}_{0}n$
   - c) $^{1}_{0}p$
   - d) $^{1}_{1}H$
   - e) $^{2}_{1}H$

6. What is the unit of the Compton wavelength?
   - a) sec
   - b) sec^{-1}
   - c) min^{-1}
   - d) m
   - e) Bq

7. To happen the photoelectric effect, what condition be essential between incident photon energy (E) and work function (W0)?
   - a) $E \geq W_0$
   - b) $E \leq W_0$
   - c) $E < W_0$
   - d) $E << W_0$
   - e) $E >> W_0$

8. What will be the length of the KYAU playground of dimension 150×80 with respect to a space shuttle crosses along of its width at speed 0.6c?
   - a) 150
   - b) 100
   - c) 80
   - d) 60
   - e) 50

9. What is the unit of Decay Constant?
   - a) Joule
   - b) Year
   - c) m
   - d) Sec^{-1}
   - e) Volt

10. A train moving with a constant speed is an example of ………………….
    - a) 1^{st} Postulate of special theory of relativity.
    - b) inertial frame of references.
    - c) 2^{nd} Postulate of special theory of relativity.
    - d) non-inertial frame of references.
    - e) Lorentz transformation.

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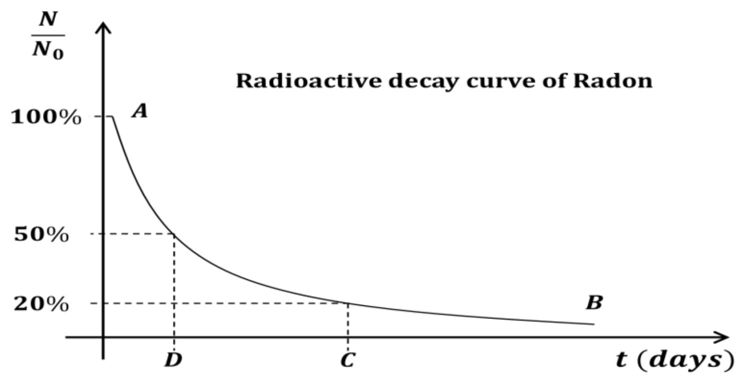
Part-B

[Answer any five (05) questions out of six (06)] **5 × 6 = 30**

1. See the stem given below and answer the following questions:



a) What is the value of point D. 1

b) Find the value of point C. 3

c) Is it possible to touch the point B with X axis? – Justify your answer. 2

2. a) Define stopping potential. 1

b) Draw the kinetic energy of photo electron vs frequency of incident light curve. 2

c) A photoelectric surface has a work function of 4 eV. What is the maximum velocity of Photoelectrons emitted by light of frequency 10^{15} Hz incident on the surface? 3

3. a) What is the mirror nucleus? 1

b) Draw the binding energy per nucleon curve. 2

c) Calculate the radius of ${}^{27}_{13}Al$ nucleus. 3

4. a) Show by the theory of relativity that the velocity of a body cannot be more than the velocity of light. 2

b) The total energy of a particle is exactly twice of its rest mass energy. Calculate its speed. 4

5. a) Define wave front. 2

b) What are the conditions of interference of light? 2

c) Point out the basic differences between two types of diffractions. 2

6. a) "For the polarizing angle of incidence, the reflected & refracted rays are 90^0 apart" – explain the statement in according with Brewster law. 3

b) The refractive index for a plastic is 1.25. calculate the angle of refraction for a ray of light incident at polarizing angle. 3